

Identification of Asali (Genuine) or Nakali (Counterfeit) Items

Project Report Part A

Information Integration and Application (CSE656)

Submitted By

Nil Makvana (MT25032)
Arpita Debnath (MT25018)
Mudit Kumar (MT25073)
Ayush Jain (MT25066)

Submitted To

Dr. Mukesh Mohania



Indraprastha Institute of Information Technology, Delhi

Department of Computer Science and Engineering

Abstract

This project implements a federated data-integration system that identifies whether pharmaceutical products are **Asali** (genuine) or **Nakali** (counterfeit). The system reconciles data across four isolated entities: Manufacturer, Distributor, Vendor, and the Ministry. By discovering shared schema connections without prior shared keys, executing parallel API calls, and evaluating cross-source integrity rules, the system generates an auditable authenticity verdict and provenance timeline.

1 Scope of Work

1.1 In Scope

- **Four Independent Databases:** Designed completely in isolation with distinct naming styles and zero shared foreign keys.
- **REST APIs & Mediator Architecture:** Every database is fronted by an independent API; a mediator handles queries exclusively through these endpoints without direct database access.
- **Instance-Based Schema Matching:** Discovers matching attributes across isolated schemas based on data-value overlap.
- **Multi-Machine Deployment:** Works across one to four machines via configuration without code changes.
- **Rule-Based Verification Engine:** Evaluates cross-source consistency to produce verdicts (ASALI, SUSPECT, NAKALI), trust scores, and provenance histories.
- **Automated Write-Back Enforcement:** Counterfeit reports submitted to the Ministry trigger an automated API call to blacklist the offending supplier at the Distributor.
- **GUI Interface:** A clean web dashboard that displays results, value conflicts, and query execution plans.

1.2 Out of Scope

Physical barcode hardware, real GS1/blockchain infrastructure, production auth/RBAC, and direct integration with live government portals.

2 New and Innovative Aspects

1. **Automated Join Key Discovery:** The system uses Jaccard set overlap on actual values to automatically identify that differing column names refer to the same item code.
2. **Federated Anomaly Detection:** Uncovers fakes by detecting inconsistencies across databases, such as cloned serials (quantity over-issue), broken custody chains, and ghost batches.
3. **Auditable Trust Scoring:** Decisions are transparent; each triggered rule applies an explicit numerical penalty or reward.

4. **Closed-Loop Enforcement:** Directly connects regulatory reporting to distributor supplier blacklisting through source-to-source API communication.
5. **Transparent Query Execution:** The GUI exposes exact sub-queries, execution times, and target sources for full inspection.
6. **Location-Transparent Mesh:** The mediator uses dynamic service discovery, running identically whether local or distributed across multiple physical machines.

3 Database Schema Design

Four isolated SQLite databases were created with distinct naming conventions and structures to replicate real-world data silos.

1. Manufacturer Database (manufacturer.db) — snake_case

- **companies** **PK** company_id | company_name | license_no | country | established_year
- **products** **PK** product_id | **FK** company_id | brand_name | generic_name | dosage_form | strength
- **manufactured_batches** **PK** batch_id | **FK** product_id | **UNIQUE** gtin_serial | mfg_date | expiry_date | batch_qty | mrp | factory_location

2. Distributor Database (distributor.db) — Abbreviated prefixes

- **suppliers** **PK** sup_id | sup_name | drug_license | gtin | region | contact_email | blacklist_flag
- **inbound_consignments** **PK** consignment_id | **FK** sup_id | source_company | received_on | invoice_no
- **distributed_stock** **PK** dist_row_id | **FK** consignment_id | item_code | product_desc | qty_supplied | unit_price | supplied_to_vendor | dispatch_date

3. Vendor Database (vendor.db) — camelCase

- **vendors** **PK** vendorId | vendorName | shopLicense | city | pincode
- **purchaseScans** **PK** scanId | **FK** vendorId | productBarcode | scannedLabelName | packSize | sellingPrice | scanTimestamp
- **customerSales** **PK** saleId | **FK** scanId | saleTime | billNo | customerPhoneHash

4. Ministry Database (ministry.db) — Business reference keys

- **counterfeit_reports** **PK** report_ref | suspect_code | reported_by_vendor | reported_against_supplier | complaint_date | lab_result | status
- **verified_genuine_registry** **PK** reg_id | **UNIQUE** auth_code | verifying_authority | verified_on

- **enforcement_actions** **PK** action_id | **FK** report_ref | action_taken | penalty_amount | action_date

3.1 Cross-Source Attribute Alignments

The table below illustrates the semantic mappings identified across the isolated schemas.

Concept	Manufacturer	Distributor	Vendor	Ministry
Item unique code	gtin_serial	item_code	productBarcode	suspect_code / auth_code
Company name	company_name	source_company	—	reported_against_supplier
Supplier name	—	sup_name	—	reported_against_supplier
Vendor name	—	supplied_to_vendor	vendorName	reported_by_vendor
Price	mrp	unit_price	sellingPrice	—

4 Populating the Data

The databases are populated with comprehensive test records: 3 companies, 10 products, 10 batches, 3 suppliers, 5 consignments, 11 distribution rows, 3 vendors, 13 scans, over 140 sales, and regulatory records. Twelve explicit validation scenarios test the rule engine:

Sample Code	Verdict	Integrity Rule Evaluated
SLP-AMOX500-B2401-000	ASALI	Verified genuine; consistent end-to-end supply chain
SLP-PANTO40-B2402-002	NAKALI	Confirmed counterfeit by Ministry laboratory testing
MRC-DOLO650-B2403-004	NAKALI	Chain of custody broken (dispatched to CityCare, sold at QuickMeds)
SLP-VITD3-B2404-0050	NAKALI	Cloned serial code: batch size 50, but 105 units scanned and sold
APX-COUGH100-B2311-00	NAKALI	Expired stock: unit sold past labeled expiry date
SLP-INSUL-B2406-0033	SUSPECT	Origin mismatch: distributor source company differs from manufacturer
ZZZ-CIPRO500-FAKE-900	NAKALI	Ghost batch: serial number absent from manufacturer records

5 APIs Creation and Schema Matching

5.1 REST API Microservices

Each database is wrapped in a dedicated FastAPI service exposing standardized endpoints:

- GET /schema and GET /sample: Return metadata and distinct sample values for schema discovery.
- POST /query: Executes parameterised, injection-safe read queries returning data and

executed SQL.

- `POST /reports` & `POST /flag-supplier`: Support write-back operations for dispute logging and supplier blacklisting.

5.2 Instance-Based Schema Matching

Column equivalence is calculated using a weighted similarity metric combining structure and content:

$$\text{score} = 0.30 \cdot \text{name_similarity} + 0.10 \cdot \text{type_similarity} + 0.60 \cdot \text{value_similarity}$$

Value similarity evaluates the Jaccard overlap of sample values retrieved via `GET /sample`. This heavy weighting allows the matcher to identify that `gtin_serial`, `item_code`, `productBarcode`, and `suspect_code` represent the same global join key.

6 Query Decomposition and Federation

The mediator maintains zero local storage. When querying an item, it performs parallel decomposed fetches using a thread pool:

- **Round 1:** Dispatches filtered queries for the given item code concurrently to all four services.
- **Round 2:** Uses primary/foreign keys returned from Round 1 to retrieve batch, consignment, vendor, and report records.
- **Round 3:** Fetches high-level parent company and supplier profile details.
- **Synthesis:** Joins all records in Python to form a complete provenance history, evaluate integrity rules, and generate the final verdict.

7 Communication Between Data Sources

All inter-service traffic uses JSON over HTTP.

- **Network Topology:** Supports running all services on a single host or distributed across two to four physical machines without modifying code.
- **Service Discovery:** Handled via `sources.json` and the `service_url()` resolver, falling back cleanly to local defaults when co-located.
- **Direct Source-to-Source Enforcement:** When a counterfeit report is logged at the Ministry service, it autonomously issues an HTTP call directly to the Distributor API to set `blacklist_flag = 1` on the supplier.
- **Fault Handling:** Network failures return clear HTTP 502 error structures rather than unhandled server crashes.

8 Query Results Integration and GUI

The mediator hosts a web UI delivering complete transparency into the integration pipeline:

- **Complete History View:** Displays an immediate status banner (ASALI, SUSPECT, or NAKALI), numerical trust score, active red flags, and an end-to-end timeline color-coded by originating data source.
- **Cross-Source Discrepancy Tracking:** Delineates between normal price variations (whole-sale vs. retail) and semantic conflicts (mismatched supplier/manufacturer).
- **Interactive Query Planner:** Displays the exact decomposition plan, including target host, SQL statement, and network latency for each sub-query.
- **Integration Lab:** Visualizes the schema matching similarity matrix, discovered column clusters, and GAV view mappings.
- **Direct Enforcement Trigger:** Enables users to file counterfeit reports with one click, updating the regulatory registry and blacklisting suppliers in real time.